

**New trading
relationships can
change the
dominant vehicle.**



**Centre for Blockchain
Technologies**

The Making of Dominant Vehicle Currencies

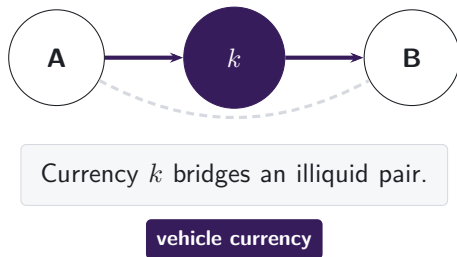
Evidence from DeFi
Jiahua Xu

A cross-border payment needs someone to make the FX market

THE LIQUIDITY-PROVIDER PROBLEM

A payment provider receives one currency but must deliver another it does not hold.

Project Nexus requires an FX provider.
Project Rialto uses a vehicle when direct exchange is unavailable.



Winning the middle concentrates trading and liquidity in the vehicle.

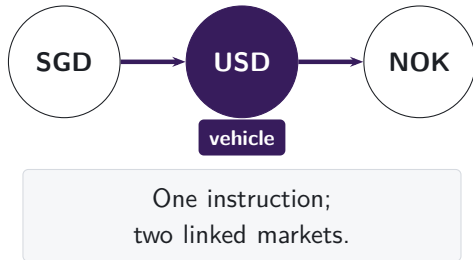
Note: Project Nexus states that every cross-border, cross-currency payment needs an actor willing and able to exchange the currencies. Project Rialto experimentally uses a vehicle currency when direct exchange is unavailable. We study the intermediary asset observed inside DEX pool routes.

Pool routes reveal the vehicle currency

SINGAPORE PAYMENT ANALOGY

A Singapore-dollar payment to Norway may be converted through US dollars.

Conventional turnover data record two legs.
Pool logs preserve their connected route.



DeFi is on-chain finance executed by smart contracts. Decentralised exchanges trade tokens through liquidity pools; **stablecoins** target a reference currency, usually the U.S. dollar.

Note: The currency labels illustrate a possible indirect conversion. Conventional FX records report turnover in each currency pair but generally do not link the two legs to one instruction. The DEX setting preserves the transaction-level pool route, so the intermediary asset is observed directly.

One transaction reveals the connected route

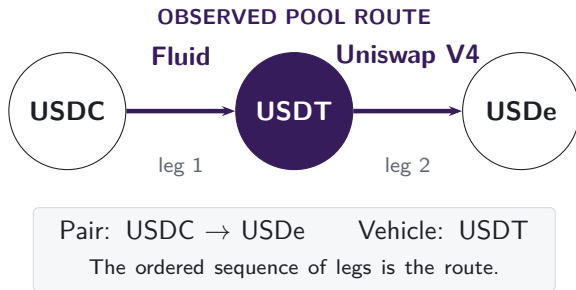
ROUTE PANEL

475 million
pool-level swaps

**November
2018–June
2026**

9
DEX deployments

Note: The route is the connected pool-swap component of transaction 0xbda1d07f...9bf67ad9 on 10 January 2026. The panel covers 2,798 dates across Uniswap v1, v2, v3, and v4; SushiSwap v2 and v3; Curve; Balancer; and Fluid. Pairs, legs, and routes are directed. The exact exchange registry maps every observed Uniswap v1 exchange to its token.



V2 turns a mandate into a market choice

V1: NOVEMBER 2018



217,003
of token-to-token routes
pass through ETH

V2: MAY 2020

95.5%

of single-leg trades use a WETH pool
in 2026

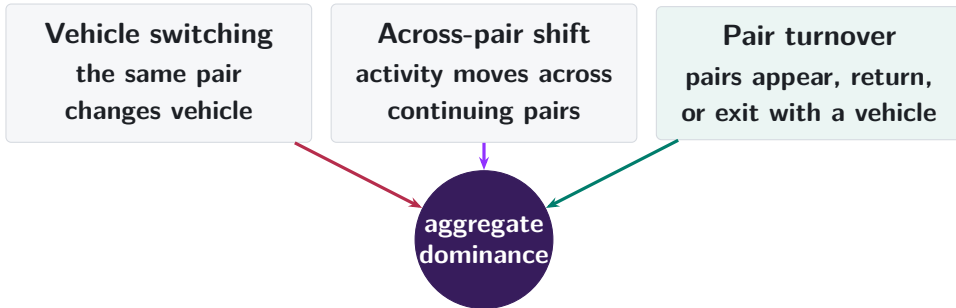
97.9%

of new token combinations include
WETH

The rule disappears; inherited liquidity can remain.

Note: V1 links token-to-ETH and ETH-to-token swaps within one transaction. V2 permits any token combination. Early V2 inherited liquidity, users, and routing conventions from V1; the 2026 comparisons show how much native-asset pairing remains after the mandate ends.

Aggregate dominance can change in three ways

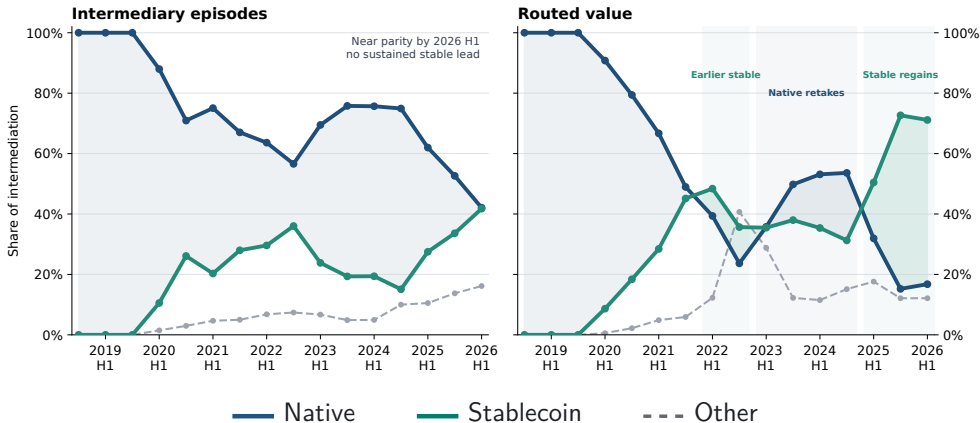


The decomposition locates the rotation; the next tests explain persistence.

Note: The decomposition is exact. All-route shares first establish the aggregate rotation. Exact two-leg routes then isolate one vehicle family per route for the pair decomposition and the stablecoin-versus-WETH comparison.

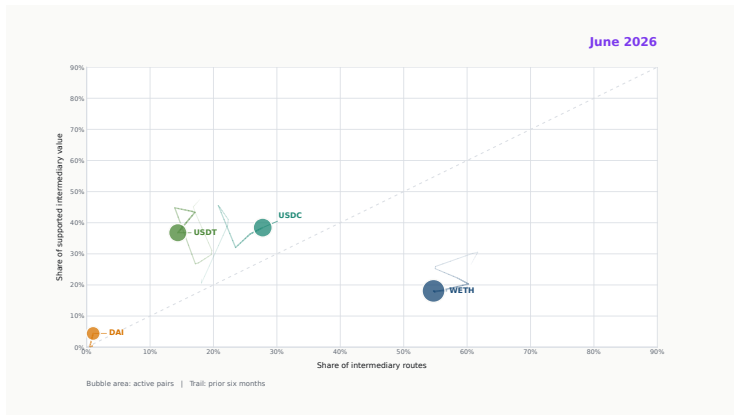
Stablecoins regain the routed-value lead by 2026 H1

ALL ROUTE LENGTHS



Note: Each complete route contributes once for every intermediary position, including routes with more than two legs. Every point covers six calendar months; the final point is 2026 H1. Routed value requires source, intermediary, and destination values to agree within 20%.

Vehicle leadership turns over through time



Click the film to play in Adobe Acrobat
Preview and browser readers show the static poster.

Note: The film follows WETH, USDC, USDT, and DAI in route participation, supported value, and pair breadth. Each frame retains six months of history. The MP4 is embedded in the final PDF and retained separately as a talk asset.

Pair formation supplies most of the rotation

EXACT DECOMPOSITION IN PERCENTAGE POINTS (PP): 2024 H1 TO 2026 H1

Same continuing
pair

−0.1 pp

Activity shifts
across pairs

+8.4 pp

First-observed
pair entry

+20.1 pp

Other lifecycle terms: −2.3 pp; net: +17.8 pp

Stablecoin
share

2024 H1
16.9% → 2026 H1
42.4%

The aggregate rises +25.5 pp; net switching inside continuing pairs is −0.1 pp.

Note: Exact two-leg native-or-stable vehicle routes on matched January–June dates. Full pair histories split first-observed entry from the period-specific net; reactivation, vehicle-role turnover, and exit form the offset. Continuing-pair weight contributes −0.5 pp. Unrounded components sum to the aggregate change.

A pair carries its first vehicle into later trading

EFFECT OF 10 PP MORE STABLECOIN USE ON THE ENTRY DAY



PAIRS TRADING AGAIN, DAYS 1-30

19.4%

PAIRS TRADING AGAIN, DAYS 31-120

12.3%

Vehicle identity persists among pairs that remain active.

Note: The sample contains 157,262 pairs entering by March 2 in 2024 or 2026. The entry day is excluded; follow-up windows do not overlap. The share regressions condition on trading again and control for cohort year, stablecoin endpoints, entry activity, direct-route share, and route complexity. Standard errors cluster by entry date: 0.15 pp and 0.18 pp, respectively.

A supported bridge becomes usable as both legs deepen

PERSISTENT SUPPORT, RELATIVE WEAK-LEG DEPTH, AND FIRST USE

FIRST STABLECOIN USE WITHIN 30 DAYS



WEEK BEFORE FIRST USE



Pool existence opens a route; two-leg depth makes it usable.

Note: Persistent support means positive prior-calendar capital on both legs on at least 24 of the next 30 days. First-use rates follow 865 pair events. The capital path uses 246 later users from day -7 to day zero; the matched change relative to WETH is +0.78 log points. First-use timing is endogenous.

Current prices can overturn incumbency

SAME PAIR, INPUT, PRETRADE STATE, AND VENUE SET

INCUMBENT RETURNS
MORE OUTPUT

93.3%

retain the incumbent



CHALLENGER RETURNS
MORE OUTPUT

27.2%

retain the incumbent

Pair and date fixed effects: challenger price leadership changes incumbent retention
by **—58.08 pp.**

Note: We quote the best feasible stablecoin and WETH routes before each observed trade. Both routes use exact Uniswap v2, SushiSwap v2, or Uniswap v3 state and keep every leg below 5% own-price impact. Price leadership requires more than 1 bp. The fixed-effect estimate has standard error 2.82 pp over 34,439 routes; inference clusters by pair and date.

Bridge capital predicts retention beyond current output

JOINT REGRESSION WITH PAIR AND DATE FIXED EFFECTS

+100 BP INCUMBENT OUTPUT
ADVANTAGE

+10.13 pp
incumbent retention

+10 PP PRIOR WEAK-LEG
CAPITAL SHARE

+2.77 pp
incumbent retention

Current exact output and prior deposited capital enter together.

Pair effects absorb persistent pair differences;

date effects absorb market-wide conditions.

Note: The outcome is incumbent retention for mature pairs with one exclusive entry vehicle family. Weak-leg capital is the smaller prior-day deposited-capital stock across the two required legs, aggregated over Uniswap v2 and SushiSwap v2. Both vehicle families have positive capital. The model contains 17,778 routes; two-way clustered standard errors are 0.77 pp and 0.61 pp.

Output shortfalls concentrate in younger pairs

EXACT STABLECOIN-VERSUS-WETH COMPARISON; INPUT-VALUE WEIGHTS

ROUTES USING THE LOWER-OUTPUT FAMILY

12.9%

Conditional median: **27.2 bp**

VALUE-WEIGHTED SHORTFALL, ALL ROUTES

7.4 bp

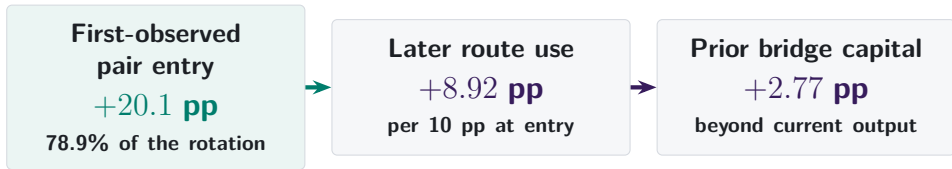
before gas



Mature pair routes track public prices more closely.

Note: A shortfall occurs when the observed family returns at least 1 bp less than the best feasible alternative family. The conditional 90th percentile is 171.5 bp. Age groups are mutually exclusive and use the same routes where both vehicle families are feasible. Values are gross of gas and cover the declared exact venues.

Dominance grows when new pairs choose a vehicle



Two margins of currency competition

Which vehicle a pair builds around; which survives price and depth changes.

Vehicle liquidity links otherwise unrelated pairs.

Note: Other lifecycle terms reduce entry's +20.1 pp to +17.8 pp net. Among retrading pairs, 10 pp more entry-day stablecoin use predicts +8.92 pp in days 1–30.

Appendix map

Definitions

- A1 Routes
- A2 Assets and pegs
- A3 Samples

Validation

- A4 Pretrade state
- A5 Pool pricing
- A6 Uniswap v1

Robustness

- A7 Timing and venue
- A8 Weights
- A9 Endpoints

Markets

- A10 Liquidity quantities
- A11 Coordination
- A12–A13 References

Further results

- A14 Network position
- A15 Endpoint demand
- A16–A17b Composition
- A18–A23 Depth and prices
- A24 Persistence robustness
- A25 LP risk

Pool data may start after the user instruction

USER INSTRUCTION RECORDED BY THE EXPLORER

Transaction details # 0x

⚡ Swap 460.00K PYUSD for 460.10K USDe on 0x 0x

THE DECISIVE SETTLEMENT TRANSFERS

USDC ERC-20 Transfer Market Maker: 0x51c7..... → Mainnetsettlr

USDC ERC-20 Transfer Mainnetsettlr → Fluid: Liquidity

The maker supplies the USDC used by the Fluid leg. The instruction is PYUSD → USDe.

Note: Transaction 0xbda1d07f...9bf67ad9, 10 January 2026. The data do not reveal whether PYUSD–USDC is another vehicle leg or executor inventory. Disconnected groups comprise 6.2% of reconstructed components in 2026. Treating each internally valid component as a separate route raises stablecoin count dominance from 42.1% to 43.7%.

A1. One reconstructed route is one unit



one route unit
two swap legs

- Each linked $i \rightarrow k \rightarrow o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k .

How we reconstruct a route from one transaction

Direct swap

one transaction



Sequential vehicle route

one transaction



Parallel split

one transaction



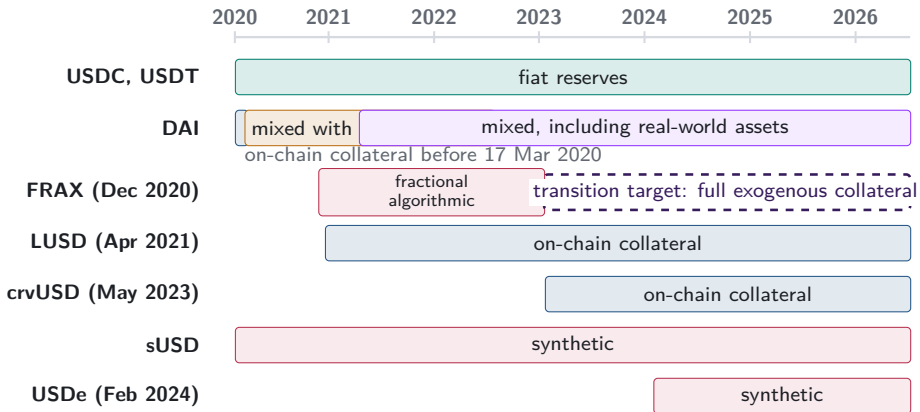
Round trip

one transaction



Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level routes reveal that choice inside aggregate turnover.

A2. Stablecoin backing changes over time



Note: Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

Peg parity preserves currency identity

Bahamian dollar $\longleftrightarrow$ **U.S. dollar**

one-for-one peg

Separate currency, monetary arrangements,
and exchange controls

USDC $\longleftrightarrow$ **USDT** $\longleftrightarrow$ **DAI**

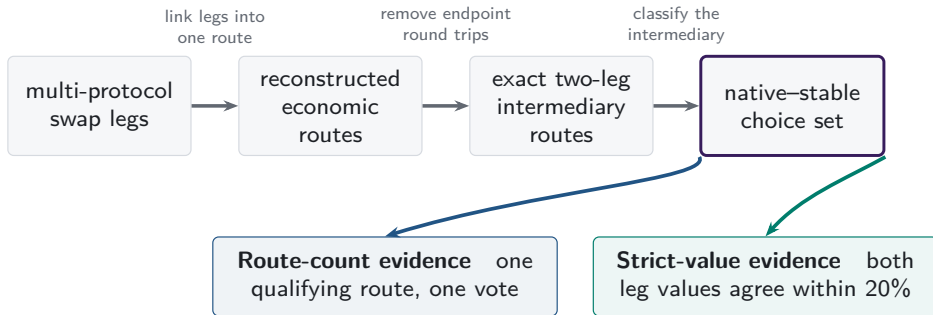
dollar target

Separate issuers, redemption, peg risk, and
pool liquidity

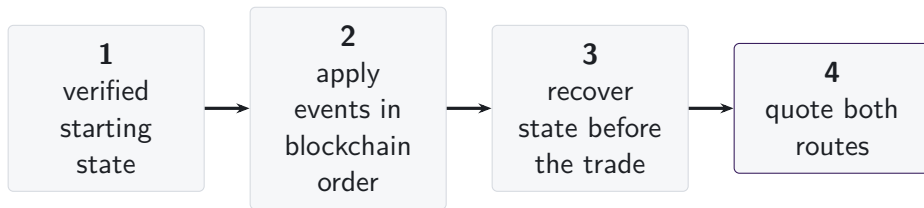
Lower divergence loss may attract liquidity providers; issuers or affiliated market makers may also seed conversion pools. Peg stress reverses the comparison.

Note: The Bahamian-dollar example follows the IMF's 2025 Article IV description of the conventional one-for-one U.S.-dollar peg. The supply-side sentence gives economic interpretations. Separating independent from issuer-linked liquidity and measuring provider returns would require provider identities, funding relationships, entry prices, fees, hedges, and expectations.

A3. One route universe supports two measurements



A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hook-dependent pricing
Constant product	■	—	—	—	—	■	—
Concentrated liquidity	■	■	—	—	—	■	—
StableSwap families	■	—	■	—	■	■	—
Weighted pools	■	—	—	■	■	■	—
V4 singleton	■	■	—	—	—	■	□

Executable-quote validation

■ recovered pricing input

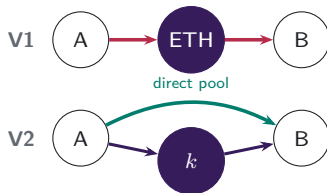
□ hook-dependent pricing unavailable

Protocol mechanisms studied

V4 singleton flash accounting and netting remain in scope; A5.3 shows the settlement mechanism.

A5.1. Pool formation determines available paths

V1 PROTOCOL RULE $\rightarrow$ V2 MARKET CHOICE



V1 admits only ETH-token pools. V2 permits any ERC-20 pool, so market creation determines whether direct and vehicle paths coexist.

Which paths exist?

Economic implication: pool creation changes the feasible route set.

A5.2. Executable depth is path-specific

V3 CONCENTRATED-LIQUIDITY POSITIONS



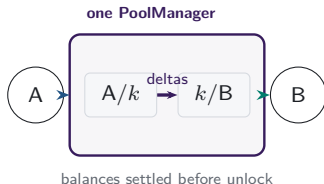
LPs choose a price range and fee tier for each atomic pool.
Positions covering the current price determine active liquidity
in the direct pool and each vehicle spoke.

Which path offers depth at this trade size?

Economic implication: executable depth is path- and size-specific.

A5.3. Shared accounting moves route settlement

V4 SINGLETON AND LOCK ACCOUNTING



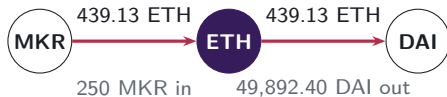
One PoolManager holds all pools. During a lock, the two route legs update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication: shared accounting changes where route obligations settle.

A6. V1 forced routes have an on-chain signature

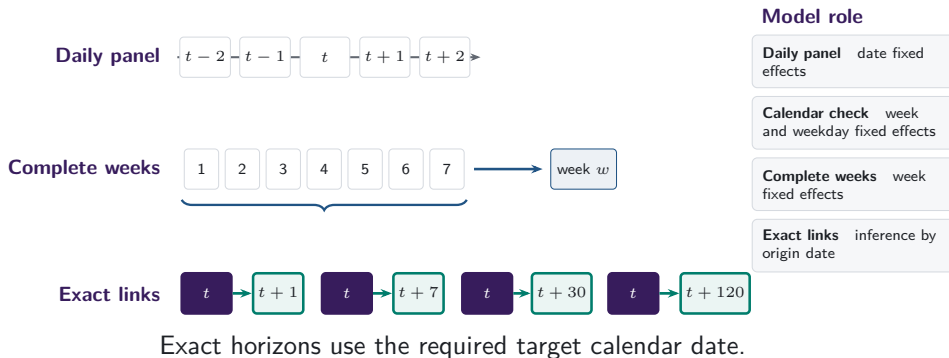
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

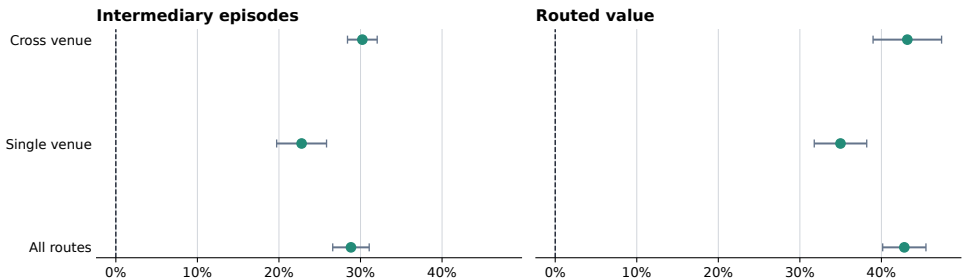
Note: Both legs of transaction 0x4dca160d...96a0ca16 (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

A7. Daily and weekly frequencies answer different questions



A7b. Stablecoin share rises within each venue scope

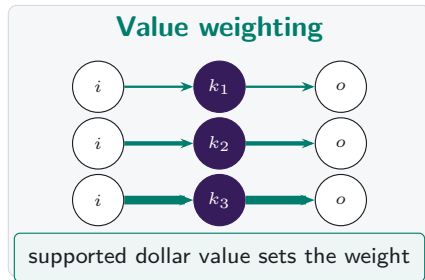
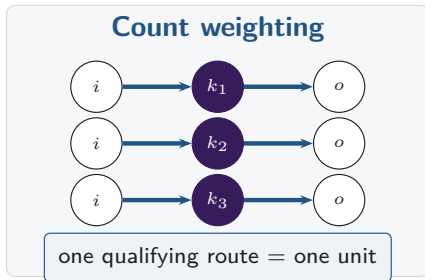
Stable share rises within realised route scopes



Points are 2024-to-2026 changes; bars are 95% HAC intervals.

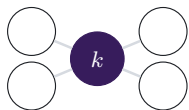
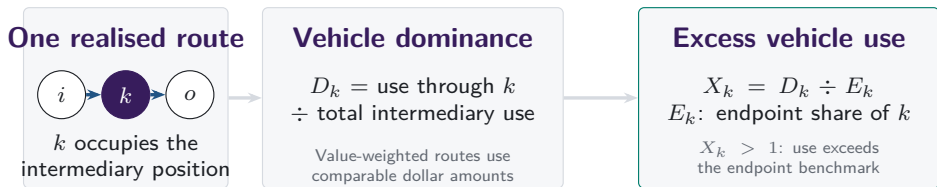
Note: Changes compare matched January-through-June dates in 2024 and 2026. Bars show calendar-HAC confidence intervals; venue scope records realised routes.

A8. Count versus value weighting

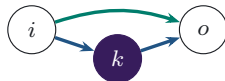


Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

Vehicle dominance aggregates realised intermediary choices



Network position



Execution cost

Endpoint use is the benchmark; network position and same-state cost remain separate.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I_{k,t}^N = \frac{n_{k,t}^I}{\sum_j n_{j,t}^I}$$

How often asset k sits inside a route.

Endpoint share

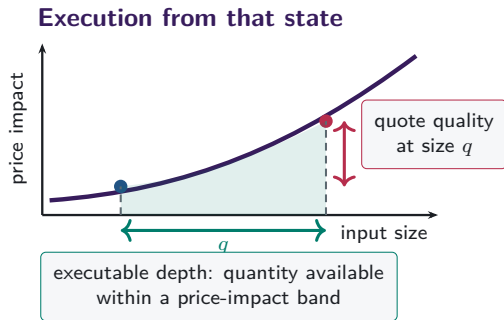
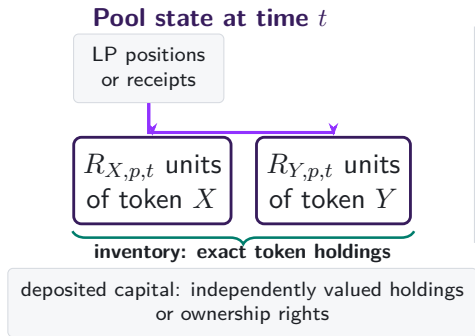
$$E_{k,t}^N = \frac{n_{k,t}^E}{\sum_j n_{j,t}^E}$$

How often k appears at either observed route endpoint.

Observed pool-route endpoints supply the benchmark.

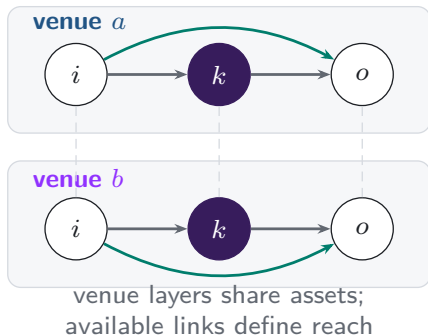
Excess use compares intermediation with observed route-endpoint use on the same support.

A10. Capital, inventory, and depth differ



A11. Additional venues expand the feasible route set

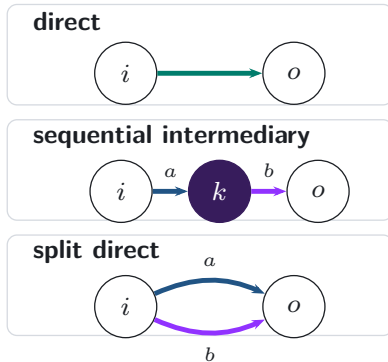
Feasible venue layers



realised
execution

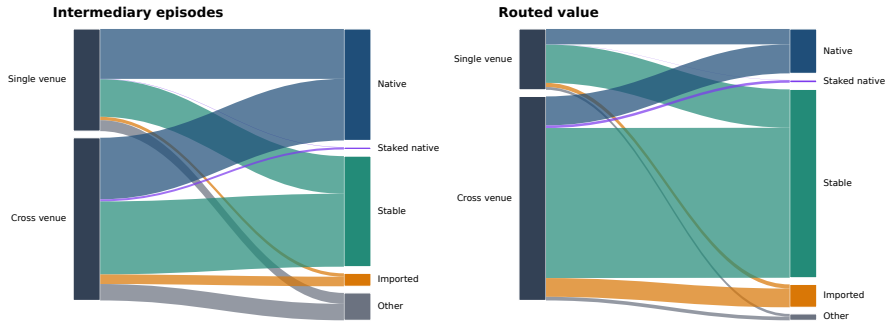


Realised route topology



A11b. Venue scope and vehicle type in 2026

Integration and vehicle composition are separate margins in 2026



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Adams et al. (2024), *Uniswap v4 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Millionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage

A14. WETH stays the graph hub as realised use shifts

ALL ROUTE LENGTHS; MONTHLY TRADING GRAPHS

WETH BETWEENNESS

#1

in all 8 annual graphs

0.925

in 2026 H1

WETH REALISED SHARE

76.2%

2024

42.4%

2026 H1

USDC + USDT REALISED
SHARE

14.9%

2024

37.2%

2026 H1

Binary network position changes much less than actual intermediary use.

Note: Betweenness is approximate and unweighted in annual trading graphs built from trades on the fifteenth day of each month, using 256 source nodes and a fixed seed. Realised shares use all intermediary positions on the same dates. A graph edge records an observed leg, not its depth or execution cost.

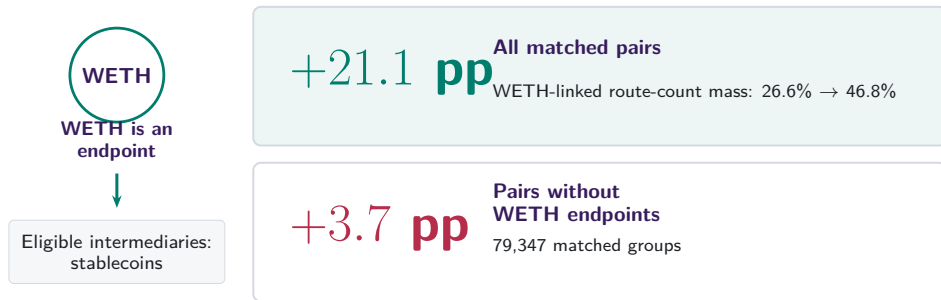
A15. Endpoint demand predicts intermediary use

	Intermediary share ($I_{a,t}$) [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	+34.56 (23.04)	+34.55 (23.04)	+0.05 (0.56)	—
Stablecoin	+2.42* (1.32)	+2.42* (1.32)	+0.85* (0.50)	—
Endpoint-demand share ($D_{a,t}$)	— —	— —	+1.59*** (0.03)	+0.98*** (0.30)
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

After date and currency effects, endpoint demand passes through almost one for one into intermediary use.

Note: 1,828,862 currency-days over 2,259 dates. Parentheses report currency/date-clustered standard errors. The outcome and demand regressor are in pp. Currency effects absorb time-invariant asset-class indicators. Asterisks *, **, and *** denote significance at 10%, 5%, and 1%, respectively.

A16. Pair composition contains a mechanical component



Changes within fixed pairs are +0.2 pp overall and +0.3 pp after removing WETH endpoints.

Note: WETH cannot be both a pair endpoint and the intermediary. Exact two-leg native-WETH-versus-stablecoin routes on common month-days; fixed-pair estimates also absorb month-day and route scope.

A17. Endpoint direction splits count and value

CONTRIBUTION TO THE STABLECOIN-SHARE INCREASE

Route count: +25.4 pp

Other pairs  +15.2 pp

One native, one stable  +9.2 pp

Two stable endpoints  +1.0 pp

Routed value: +43.9 pp

Other pairs  +20.6 pp

One native, one stable  +10.1 pp

Two stable endpoints  +13.2 pp

USDT supplies +13.7 pp of the +13.2 pp two-stable-endpoint value channel.

Note: Exact additive decomposition on 181 matched dates in 2024 and 2026. Endpoint direction describes the pair; the vehicle remains the intermediary token.

A17b. Rotation survives time and endpoint restrictions

**-0.4 pp to
+1.0 pp**
within-pair term across
all adjacent H1
comparisons

**1.1% →
9.0%**
route-count share with
nonvehicle endpoints

**0.9% →
39.1%**
supported-value share with
nonvehicle endpoints

Pair entry and exit contribute +7.2 pp by count and +33.8 pp by value.

Note: Adjacent comparisons use matched January–June dates from 2019 through 2026. The endpoint restriction removes every pair with WETH or a stablecoin at either endpoint. All changes and components are in pp; supported value applies the 20% agreement rule.

A18. Local depth remains informative alongside network reach

84.1%

of routes use the deepest supported bridge

+6.79 pp

route share per log point of weak-leg capital

+9.10 pp

route share per log point of same-day network reach

Local two-leg depth and broader network position both matter.

Note: Five vehicle currencies compared within the same set defined by pair, date, and route scope. Depth is prior-calendar deposited capital on the weaker leg. The regression absorbs set and vehicle effects and clusters by pair and date.

A19. First-use capital lies in active pools

SUPPORTED STABLECOIN, DAY -7 TO FIRST USE

92.5%

of first-use capital lies in
continuing pools

+0.16

continuing-pool growth
relative to WETH

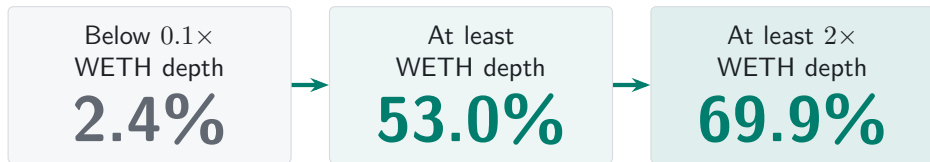
+6.9 pp

greater probability of a
newly active pool

Note: 246 first-use events. A newly active pool has positive prior-calendar capital on day zero and none on day -7; continuing pools are positive on both dates. Difference inference clusters by pair and adoption date.

A20. Relative depth predicts route flow

FIRST 30 DAYS AFTER PERSISTENT STABLECOIN SUPPORT



Vehicle competition turns on relative depth.

Note: Depth is prior-calendar deposited capital on the weaker leg. The route-share comparison uses 11,327 pair-days across 855 bridge events. Support and depth refer to the same stablecoin set. Capital and route demand are jointly determined.

A21. Usable depth turns support into first use

PROBABILITY OF FIRST STABLECOIN USE WITHIN 30 DAYS

BELOW $0.1 \times$ WETH DEPTH

42.6%



AT LEAST $0.1 \times$ WETH
DEPTH

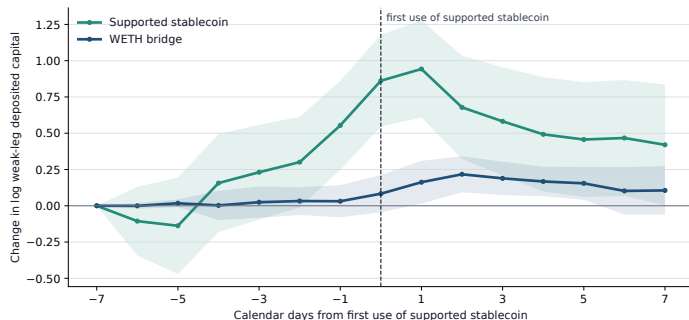
84.1%

Positive pool capital becomes useful as both route legs deepen.

Note: The sample follows 865 pair events after persistent support appears. The estimates compare first use within 30 days. Lower divergence loss and issuer-linked pool seeding are two possible supply channels. Provider profitability also depends on entry prices, fee income, hedges, and expectations.

A22. Bridge capital builds before first use

SUPPORTED STABLECOIN FIRST USED AT DAY ZERO



BEFORE FIRST USE

+0.86

stablecoin bridge, day -7
to 0

AFTER FIRST USE

-0.44

stablecoin bridge, day 0
to +7

Capital builds into first use, then partly unwinds.

Note: 246 pair events with first use 7 to 119 days after persistent support. Relative to WETH, capital changes +0.78 before and -0.46 after first use. 92.5% of first-use capital lies in pools active one week earlier. Capital is lagged one day; intervals cluster by pair and adoption date.

A23. Broader price search mostly changes the venue

SHARE OF ROUTES WITH A QUOTE MORE THAN 1 BP BETTER THAN EXECUTION

SAME VEHICLE,
USED VENUES

6.6%



SAME VEHICLE, ALL
EXACT VENUES

44.5%



ANY VEHICLE OR
DIRECT

46.4%

Opening other vehicles adds 2.0 pp; stablecoin share changes by -1.2 pp.

Note: Routes are repriced for the same pair and input immediately before execution. The sample contains 777,203 routes on 73 monthly dates across exact Uniswap v2, SushiSwap v2, and Uniswap v3 states; minimum input is \$100 and the impact limit is 5%. Quotes are gross of gas.

A24. Persistence is equally strong on busy entry days

EFFECT OF 10 PP MORE STABLECOIN USE ON THE ENTRY DAY

	Days 1–30	Days 31–120
At least 5 entry routes	+9.70 pp	+9.31 pp
At least 10 entry routes	+9.65 pp	+9.35 pp

Stronger entry measurement leaves the persistence estimate intact.

Note: The entry day is excluded and the follow-up windows do not overlap. The 5-route samples contain 1,629 and 885 pairs; the 10-route samples contain 671 and 452. Controls match the main pair-weighted specification, and standard errors cluster by entry date.

A25. Divergence risk helps locate bridge capital

EFFECT OF 10 PP MORE PRIOR RELATIVE VOLATILITY

LOG BRIDGE CAPITAL

—0.117

current date (0.049)

—0.093

30 days later (0.029)

MEDIAN RELATIVE
VOLATILITY

Native 126.9%

Stablecoin 148.4%

Stablecoin bridge has lower risk in
29.4% of pair-months.

Risk shapes local capital; the rotation extends beyond it.

Note: The panel contains 58,447 constant-product WETH and stablecoin bridge observations with positive capital on both legs. Models absorb pair-by-date and vehicle effects, control for prior route activity, and cluster by 4,254 pairs and 71 dates. Daily divergence loss gives the same directional result. The comparison describes deposited capital, not provider returns.